CIP-A106 Laboratory 3

Evidence Manifest and Findings Matrix

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1. Chronological evidence manifest

Columns: artefact, filename, SHA-256, source, time, analysis tool, and whether the item is in this package. Items marked "No" stay in the VM (live malware, raw memory and executable images are not submitted). Screenshots are listed as groups here and one per row in screenshots_manifest.csv. The same data is in evidence_manifest.csv.

Time	Artefact	Filename	SHA-256	Source	Tool	In package
2021-04-23 06:11:30	Archive (outer)	malware sample 4 archive (name as supplied)	63E52C4A52CBFB9AB1885CBB83E5D5B836802A06618B3BC07EBC20545DF75729	Course material, source archive	PowerShell Get-FileHash	No (hash only)
2021-04-23 06:11:30	Nested archive	malware+sample+4.zip	B049A5EC61205581C06C764A675983C26C2989374E22B167FD930DDC64E138CE	Extracted from outer archive in the VM	PowerShell Get-FileHash	No (hash only)
2021-04-23 06:11:30	Password file	password (7).txt	2F496C63E0BB77B76A65141982050E0D9CA887AE71B4856B4E984E0BDB4785F3	Extracted from outer archive in the VM	PowerShell Get-FileHash	No (hash only)
2021-04-23 06:11:30	Source executable (O)	original\sample\malware sample 4\demo1_ransomware.bin	5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D	Extracted from nested archive (368,640 B)	PowerShell Get-FileHash, PEStudio	No (live malware)
2026-10-03 11:40:27	Working copy (W)	work\demo1_ransomware.exe	5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D	Copy of O (identical hash)	PowerShell Get-FileHash	No (live malware)
2026-10-03 11:49-12:02	PEStudio screenshots E01-E08	E01..E08 (see SHA256SUMS.txt)	see SHA256SUMS.txt	Host screen capture of the VM	PEStudio 9.61	Yes
2026-10-03 12:04	VM isolation and snapshot screenshots	E09-E11	see SHA256SUMS.txt	Host screen capture of VMware settings	VMware Workstation	Yes
2026-10-03 12:12-12:55	x32dbg session screenshots (PIDs 7380, 8360, 7784)	E12-E19	see SHA256SUMS.txt	Host screen capture	x32dbg	Yes
2026-10-03 12:45:25	Uncontrolled first run: self-copy (D1)	C:\Windows\gudpnujaukba.exe	5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D	Created by the sample, PID 8532 (Hidden+Archive, LastWriteTime 2017-04-01 18:11:02)	PowerShell Get-FileHash	No (live malware)

Time	Artefact	Filename	SHA-256	Source	Tool	In package
2026-10-04 01:12:08	Ransom note (text)	evidence\07_note_recovery_txt.txt (253 B note recover_file_*.txt, text copy)	4EB0950CFE07D80EF245D06938BFECCBC667C8E68EBB6C8CA9700C13CD25FAE8	Note text copied from the Desktop/Documents folder	PowerShell	Yes (rebuilt text, 06_Text_Exports)
2026-10-04 01:16-01:37	Debugger and System Informer screenshots (PIDs 3848, 8532)	E20-E28	see SHA256SUMS.txt	Host screen capture	x32dbg, System Informer 3.2	Yes
2026-10-04 01:36:06	Encrypted dump of PID 8532	memory\gudpnujaukba_8532.dmp.mp3	1F9F919EBC1D6077BE971762BAC200507D87EE45FA60FCBCF1D2B95D268669CC	System Informer Create dump file, then encrypted by the sample (132,232,656 B, not a valid MDMP)	PowerShell Get-FileHash	No (raw memory, encrypted)
2026-10-04 01:50:03	Process 4820 started (copy in Documents)	C:\Users\cybersky\Documents\bbflnrupddix.exe	5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D	Started by the sample (parent shown as non-existent process 1120)	System Informer, Prefetch	No (live malware)
2026-10-04 01:58:52	Desktop notes listing	evidence\08_desktop_notes.txt	3E08ACD8555D3BFC59365E146AD2C4C0BFA9E13D30D42DA98843C973337223D1	Directory listing of the Desktop	PowerShell	Yes (rebuilt text)
2026-10-04 01:58:54	Recent lab folder listing	evidence\08_labs_recent.txt	1FD8EA03E74C319EA6707BBEAC9F15D555DB21E4D6E7B8E571C91B6AEAE6A2B4	Listing of recently changed files under C:\Labs	PowerShell	Yes (rebuilt text)
2026-10-04 02:04:28	Dump record after first run	evidence\09_dump_after.txt	DA9791EBC7AF7F5193BABF62C4B03A428D02B6979FD3DDAA2F5BC88368EE51E6	Hash, size and header of the encrypted 8532 dump	PowerShell	Yes (rebuilt text)
2026-10-04 02:06:05	.mp3 file list	evidence\09_mp3_files.txt	DC0C35E75B3ED80DE804289A0099865F3486BD6FA065B5094D3AF83F83D9BE6B	List of renamed (encrypted) files, 171,994 B	PowerShell	No (large list, retained in VM)
2026-10-04 02:16:46	Memory dump of PID 4820	memory\bbflnrupddix_4820.dmp	6FA90986C74D9383B9FC0A9CC0AD3D06A92357E15EA95D4075AFEB4BFC79C0B8	System Informer Create dump file, valid MDMP, 117,739,615 B, 357 regions, 55 modules	System Informer 3.2	No (raw memory dump)
2026-10-04 02:17:18	Dump record (4820)	evidence\10_dump_record.txt	90515C3C865ABBC5AC4516A1B4E26283BDFE477889AD35155212DE4247B40B89	Hash and header of the 4820 dump	PowerShell	Yes (rebuilt text)
2026-10-04 02:24:43	Validator output (Python, unusable)	evidence\11_dump_validation.txt	04258E28F0608950453FCD31E567939AAA177CB6BFFAB3DA35F174C20EE2BBD0	System Python broken (stdlib renamed to .mp3); output records the failure only	PowerShell	Yes (rebuilt text)
2026-10-04 02:28:03	Dump validator (C#)	work\dumpcheck.cs	299BA6B0AAD31969FE2E089C2106746544A68827869C6A62460E164AAA8DFDFC	Script written during the exercise (7,809 B)	PowerShell Add-Type	No (retained in VM)
2026-10-04 02:28:05	Unpacked image extracted from 4820 dump	unpacked\mem_image_0x400000.bin	42900DE68702ADB3E6802155AA416C7EEB5F19EBCC7D7387C08BAA4C51D7D9A8	544,768 B image at 0x400000 (executable code)	dumpcheck.cs	No (unpacked executable image)

Time	Artefact	Filename	SHA-256	Source	Tool	In package
2026-10-04 02:28:08	Validator output (C#, 4820 dump)	evidence\11_dump_validation_cs.txt	25C57BEA79F88ACE1CDB7F618FFCE77280972E88D668978AFFDA887D7C8FB65B	Modules, sections, string tables. The file-entropy and differing columns are invalid (mismatched section tables) and are not cited	dumpcheck.cs	Yes (rebuilt text)
2026-10-04 02:22-02:23	System Informer screenshots PID 4820	E30-E33	see SHA256SUMS.txt	Host screen capture	System Informer 3.2	Yes
2026-10-04 11:37:41	Payload image summary	evidence\12_payload_image.txt	E4CA308903876E30EBCAD266C4BD5CD6AD27CC2FEBD847999ABA37DBC87EF888	Section summary of the unpacked image	dumpcheck.cs	Yes (rebuilt text)
2026-10-04 11:37:45	Payload .rdata strings	evidence\12_payload_rdata_strings.txt	323580A2D31DBE380E2C1F3F159B9C034FB8897D398F5693AC2F641E7844685E	Strings from the unpacked .rdata, 68,956 B	dumpcheck.cs	No (large list, retained in VM)
2026-10-04 11:46-12:43	x32dbg sessions S1-S4 (PIDs 5348, 6068, 7152)	E34-E57	see SHA256SUMS.txt	Host screen capture	x32dbg	Yes
2026-10-04 11:58:17	OEP dump record	evidence\13_oep_dump_record.txt	6DE81DE30A142BE4C2CD9FF53221ECA742F87B64F706258527E0992B4B8B5F9E	Validation of the OEP dump (30 modules, section hashes, string tables)	dumpcheck.cs	Yes (rebuilt text)
2026-10-04 12:50:14	Memory dump at OEP (PID 7152)	memory\demo1_ransomware_7152_oep.dmp	C93F7330B0EC11CF22915D246F56BC59F51DAFDE17BE724613531358C5F192AB	Dump taken while stopped at OEP 004265E8; valid MDMP, 43,342,498 B, 218 regions, 30 modules	System Informer 3.2	No (raw memory dump)
2026-10-04 12:51:23	Unpacked image extracted at OEP	unpacked\oep\mem_image_0x400000.bin	8DE53033FFB8C5936EE7BFA7BDFFA4C15ED1F2D98C9029693E4239C4A4EA75E5	544,768 B image at 0x400000	dumpcheck.cs	No (unpacked executable image)
2026-10-04 13:00	x32dbg Log screenshot, six VirtualProtect calls	E57	see SHA256SUMS.txt	Host screen capture	x32dbg	Yes
2026-10-04 13:03:34	x32dbg log export	C:\Tools\x64dbg\release\x32\log-20261004-130334.txt	7EBA1AE9... (prefix only recorded)	x32dbg Log Save (26,491 B)	x32dbg	No (retained in VM)
2026-10-04 13:11:43	x32dbg log (clipboard copy)	evidence\14_x32dbg_log_7152.txt	6DD5916DEFBBEBB529CA8F23B1D3F2A58938AC8640384670846C027D90A0E32D	26,587 B. Lines 86-160 reproduced in 04_Logs	x32dbg, PowerShell	Excerpt only
2026-10-04 13:25:01	x32dbg log (saved copy)	evidence\14_x32dbg_log_saved_130415.txt	A593DE3D2AC395F5D8BDC38281732A1BDA85C7F2ED9815183DEF6C84B28DD5E9	26,539 B	x32dbg, PowerShell	No (retained in VM)
2026-10-04 13:30	Registry Run keys before clean-up	HKCU and HKLM ...\CurrentVersion\Run (console output)	n/a (console output)	Five malicious HKCU values, HKLM clean	PowerShell Get-ItemProperty	Yes (in report and log)
2026-10-04 13:32:40	HKCU Run export after clean-up	evidence\15_hkcu_run_after.reg	53EFE0078D0377592E11EA3FC83D75CAC3C8F57147242E777B8912582290B3FB	Exported after the five Run values were removed	reg export	Yes (rebuilt text)

Time	Artefact	Filename	SHA-256	Source	Tool	In package
2026-10-04 13:39	VM snapshot and isolation screenshots	E58-E64	see SHA256SUMS.txt	Host screen capture of VMware settings	VMware Workstation	Yes
2026-10-04 13:55:11	PowerShell transcript copy	evidence\17_transcript_obtmgjOa_20261004014859_copy.txt	0A50C6504032205B3D2DE3A3977092AB7E0FFE2A22275F4A715490F4ED7A6615	225,908 B, copy of the locked live transcript	PowerShell Copy-Item, Get-FileHash	No (retained in VM)
2026-10-04 13:56:04	PowerShell transcript copy	evidence\17_transcript_40wl0EbZ_20261004125102_copy.txt	632EC466871F2D52B622D7BA186FD64D78F390B4CC49BA4586BE8EB37B6135FF	82,266 B	PowerShell Copy-Item, Get-FileHash	No (retained in VM)
2026-10-04 14:02	Text bundle	18_text_bundle.txt	1BE0909538F49258F8F3196CA9512CE6A8571C3050309C6C12D63B1A2E5B66EC	Ten text exports joined for transfer by paste (20,383 B)	PowerShell	No (source of 06_Text_Exports)
2026-10-03 02:13 (10/4)	Encrypted baseline transcript (not recoverable)	PowerShell_transcript...20261003110249.txt.mp3	99810D8AC736DD8C74C3ACDE72D4361FD7F6E2043500E88EB449F6EF7DF8F42A	Original baseline transcript (111,012 B) encrypted by the sample at 2:13 AM on 10/4	PowerShell Get-FileHash	No (encrypted)

2. Findings matrix

Every behavioural conclusion, IOC and ATT&CK mapping with its source artefact, confidence level and limitation. Confidence: High = observed directly by a tool; Medium = observed in memory or by one source only; Low = inference or unconfirmed.

ID	Type	Finding / indicator	Source artefact	Confidence	Limitation
F01	Behaviour	Sample is packed (custom or unidentified packer)	E01-E08 PEStudio; entropy per section; 6 imports	High	Packer not named
F02	Behaviour	Stub allocates memory (VirtualAlloc x3) and restores protections (VirtualProtect x6)	E47-E51, E54, E57; x32dbg log	High	Calls 1,2 only partly in S3; call 6 from S4 screenshot
F03	Behaviour	Original entry point 004265E8 (RVA 0x265E8)	E52, E54; dump PE header	High	None across three runs
F04	Memory	Unpacked image recovered (4 sections, SizeOfImage 0x85000)	Dumps 4820 and 7152; section hashes	High	.data differs (runtime state)
F05	Memory	Victim ID and Bitcoin-format address appear only after the payload runs	13_oep_dump_record vs 11_dump_validation_cs	Medium	Two dumps; absence is not proof
F06	Behaviour	Files encrypted and renamed .mp3 with victim ID header	09_dump_after header bytes; 09_mp3_files; E29	High	Cipher and key handling not analysed
F07	Behaviour	Ransom notes dropped in many folders	File listings; 07_note_recovery_txt	High	Raw notes not packaged
F08	Behaviour	Self-copy with Hidden attribute, same hash	Get-Item and Get-FileHash (13:30)	High	LastWriteTime preserved from source
F09	Behaviour	Persistence by five HKCU Run values	Get-ItemProperty 13:30; 15_hkcu_run_after.reg	High	Three created in debug sessions S1-S3

ID	Type	Finding / indicator	Source artefact	Confidence	Limitation
F10	Behaviour	Registry key HKCU\Software\xxxsys\ID	Registry query after first run	High	EnableLinkedConnections origin unverified
F11	Behaviour	Attempt to run wmic.exe (possible shadow-copy deletion)	E18, E29 dialogs; wmic string (3 hits)	Low	Command line not captured; WMIC absent; vssadmin string 0 hits
F12	Capability	Network capability (wininet, ws2_32, domains, user-agent)	Imports and strings in memory	Medium	No traffic captured; adapter disconnected
F13	Behaviour	Relaunched copies and original exit with code 1 in debugged runs	x32dbg Log S1-S3	Low	Cause unknown (hypothesis: mutex or check)
F14	Behaviour	Threads at 0041EAA0 and 0041FEB0 exist in the running copy but breakpoints never fired when debugged	E24-E27 threads; breakpoint hit counts	Low	Purpose unknown
I01	IOC: Victim ID	4855EB75F88FBD5 (registry and file-header form 4855EB75F88FBD05)	Ransom note text; HKCU\Software\xxxsys\ID; first 16 bytes of encrypted files	High	See report section 8
I02	IOC: Bitcoin-format address	1E3Hf3Mpy4g8zZUHPoAFwSarq5UxLbocT3	Ransom note; payload memory after the run (format only, not validated on any ledger)	High (presence)	See report section 8
I03	IOC: Domain	gwe32fdr74bhfsyujb34gfszfv[.]zatcurr[.]com	Unpacked image (6 hits) and notes; URL form hxxp://<domain>/<ID>	High (presence); no traffic observed	Presence only, not contacted
I04	IOC: Domain	tes543berda73i48fsdfsd[.]keratadze[.]at	As above	High (presence); no traffic observed	Presence only, not contacted
I05	IOC: Domain	tt54rfdjhb34rfbnknaerg[.]milerteddy[.]com	As above	High (presence); no traffic observed	Presence only, not contacted
I06	IOC: Onion address	xlowfznrg4wf7dli[.]onion/<ID>	Unpacked image (3 hits) and notes	High (presence); no traffic observed	Presence only, not contacted
I07	IOC: Registry key	HKCU\Software\xxxsys (value ID)	Registry query after first run	High	See report section 8
I08	IOC: Run values (HKCU)	iorpyhcbjsja, vylvawuyxpdy, lccjvxmpqmjl, objdqnybfhvp, wikqxbughfcu, each cmd.exe /c start "" "<copy>"	Get-ItemProperty 2026-10-04 13:30	High	See report section 8
I09	IOC: File name (self-copies)	C:\Windows\gudpnujaukba.exe, vltxuvwdbsbu.exe, raknlbaoydys.exe, wlcdcfqqewfn.exe; Documents\ bbflnrupddix.exe	File listings; identical SHA-256 to the source	High	See report section 8
I10	IOC: File name (notes)	_RECOVERY_+<suffix>.txt/.html/.png, recover_file_<suffix>.txt, RECOVERY.TXT/HTM/png	File listings	High	See report section 8
I11	IOC: File extension	.mp3 appended to user files; first 8 bytes zero, then victim ID	Header bytes of encrypted files; 8,009 .mp3 files counted at 02:09 AM	High	See report section 8

ID	Type	Finding / indicator	Source artefact	Confidence	Limitation
I12	IOC: Hash (SHA-256)	5343947829609F69E84FE7E8172C38EE018EDE3C9898D4895275F596AC54320D	Source sample and all five self-copies	High	See report section 8
I13	IOC: Hash (imphash)	C00702BDB5E1419C3DC899A74A60A37D	PEStudio	High	See report section 8
I14	IOC: PDB path	E:\Tools\aolfed\release\osc.pdb	PEStudio debug directory	High	See report section 8
I15	IOC: Version strings	"nah nah Corporation", "nah nahApp", 1.600.5512	PEStudio	High (provenance only)	See report section 8
I16	IOC: Strings	pewpewpew, die hard, Joyhv.pew, nbvgg.hlp, control.ini, kasdfgh283, ityeofm9234-23423 (purpose unknown)	Strings v2.54 and unpacked image	Medium	See report section 8
I17	IOC: User-agent	Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; Touch; rv:11.0) like Gecko	Unpacked image	High (presence); no traffic observed	Presence only, not contacted
A-T1027.002	ATT&CK T1027.002	Obfuscated Files or Information: Software Packing (Observed)	Section entropy 7.86 to 7.88 (.crt, CODE), 6 imports, entry in a small stub, in-memory image differs from disk, six VirtualProtect calls then OEP hit (Ex. E01, E04, E57, E52)	High	See report section 9
A-T1486	ATT&CK T1486	Data Encrypted for Impact (Observed)	User files renamed to .mp3 with an 8-byte header holding the victim ID; notes dropped; evidence and transcripts encrypted. The encryption algorithm was not examined (Ex. E29, E28)	High	See report section 9
A-T1547.001	ATT&CK T1547.001	Boot or Logon Autostart: Registry Run Keys (Observed)	Five HKCU Run values, each cmd.exe /c start "" "<copy>" (console output 13:30)	High	See report section 9
A-T1564.001	ATT&CK T1564.001	Hide Artifacts: Hidden Files and Directories (Observed)	Self-copies carry Hidden+Archive attributes	High	See report section 9
A-T1036	ATT&CK T1036	Masquerading (Inferred)	Random self-copy names and the .mp3 extension on encrypted files. No attempt to imitate a legitimate program name was seen	Low	See report section 9
A-T1112	ATT&CK T1112	Modify Registry (Observed (xxxsys); unverified (EnableLinkedConnections))	HKCU\Software\xxxsys\ID written. The origin of EnableLinkedConnections=1 was not verified	Medium	See report section 9
A-T1490	ATT&CK T1490	Inhibit System Recovery (Attempted, not confirmed)	A "Windows cannot find wmic.exe" dialog appeared in four screenshots and the string wmic is in the payload image (3 hits). The command line was not captured and WMIC is absent on this Windows build, so no shadow-copy deletion effect is claimed (Ex. E18, E29)	Low	See report section 9